CODECHECK certificate



Reviewed paper: Egor Prikaziuk, Peiqi Yang, Christiaan van der Tol (2021): Google Earth Engine Sentinel-3 OLCI Level-1 Dataset Deviates from the Original Data: Causes and Consequences. *Remote Sensing*. 2021; 13(6):1098. https://doi.org/10.3390/rs13061098

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Summary: The reproduction was successful. The authors followed good scientific practice and provided open code via a GitHub repository and open data via DANS Easy. After some clarification with the corresponding author, all relevant figures from the paper could be reproduced with only minor deviations from the figures reported in the paper. The author demonstrated concern for the reproducibility of the work by improving the code throughout the review process.

CODECHECK Repository: https://github.com/MarkusKonk/S3_loader

CODECHECKER notes

The paper is already published and contains a data availability statement, including links to the data and the code. The entire workflow has three steps:

1. Downloading data
2. Extraction of pixels
3. Creating the figures

Downloading data

The data download is well documented in the repository's README and done using the file *example.py*. It requires some manual steps before the code can be run, i.e., registering on the website to generate the API key and completing the configuration file accordingly. The download takes a while and might need several iterations to complete. ***Recommendation:*** *Add a note to the README to make sure authors are aware of that.* Furthermore, the resulting folder with the datasets is getting large (>20GB), which is also why it not available on DANS Easy.

Extraction of pixels

After the download, the lines 62-65 in the same file need to be run, which were commented out before. ***Recommendation:*** *Create two files since users might be confused when to comment out what.* Running the code creates the folder *example_extracted*, used by the notebook that creates the figures in the paper. The folder is available on DANS. Thus, users interested in the figures only could skip the download and the extraction and use the notebook directly.

Creating the figures

The focus of the work is on the S3_loader package, i.e., the download and the extraction. However, the paper also contains a number of figures created using the notebook. I first forked the repository and ran the *requirements.txt* file. Due to an error (*Could not find a version that satisfies the requirement shapely~=1.6.4.*), I had to change the version of the package *shapely* from 1.6.4 to 1.7.1. I also added the packages *matplotlib==3.4.2* and *pandas==1.3.0* to the *requirements.txt*. Then, I ran the first chunks of the code. A message in the code informed me to download the data from DANS. I created a new folder *figures* and added the code *savefigure(...)* to each figure to store them in the folder. ***Recommendation:*** *Such a folder is not only needed for a CODECHECK to compare original and reproduced figures but also good practice, as suggested by Nüst et al. (2018).*

In the following, the reproduced figures are shown. The left figure is the reproduced one, whereas the right one is a screenshot from the paper. The comparison was made visually. According to the author, deviations are caused by the 3x3 extraction stored in DANS. In the paper, a 1x1 extraction was used.

Figure 3: While the left part has some slight deviations (see, e.g., the two highest bars), the right side seems identical.

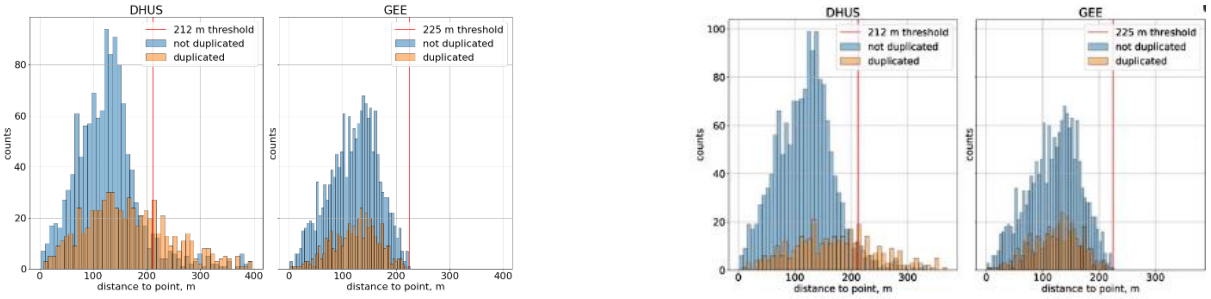


Figure 4: The deviations on the right side are likely caused by the different scales due to the deviations in the left part of the figure.

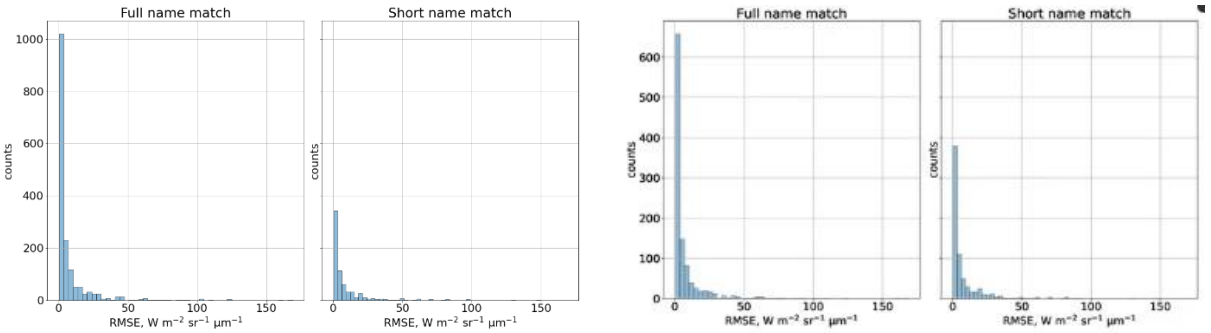


Figure 5: Figures seem to be equal.

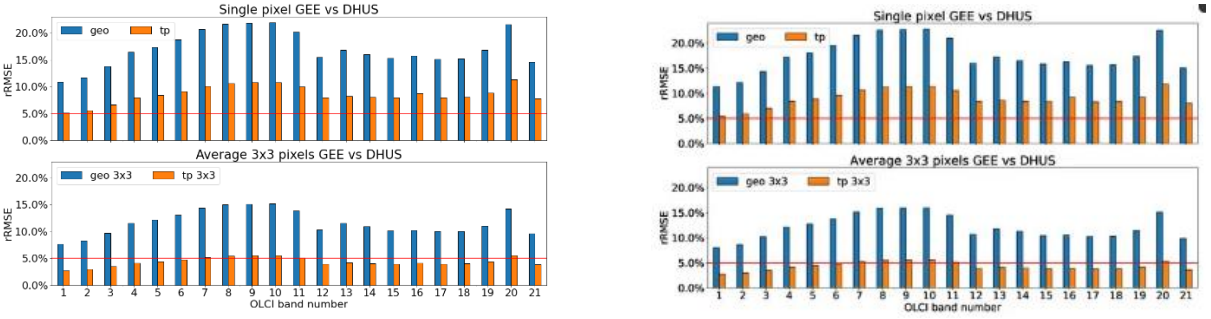


Figure 6: Figures seem to be equal.

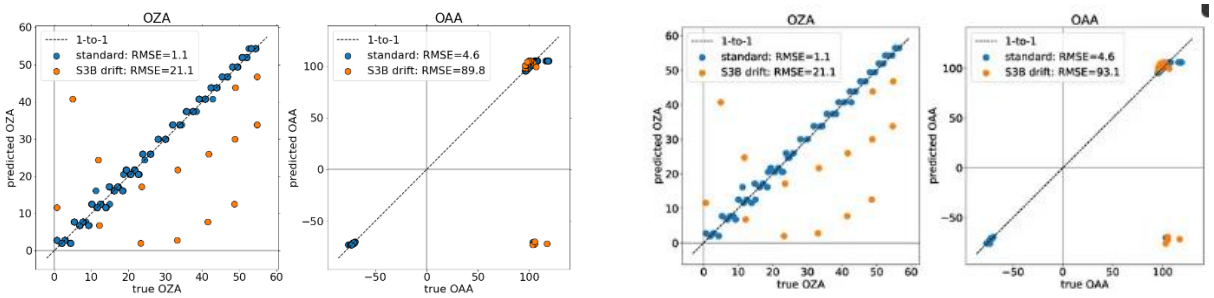


Figure 7: Figures seem to be equal.

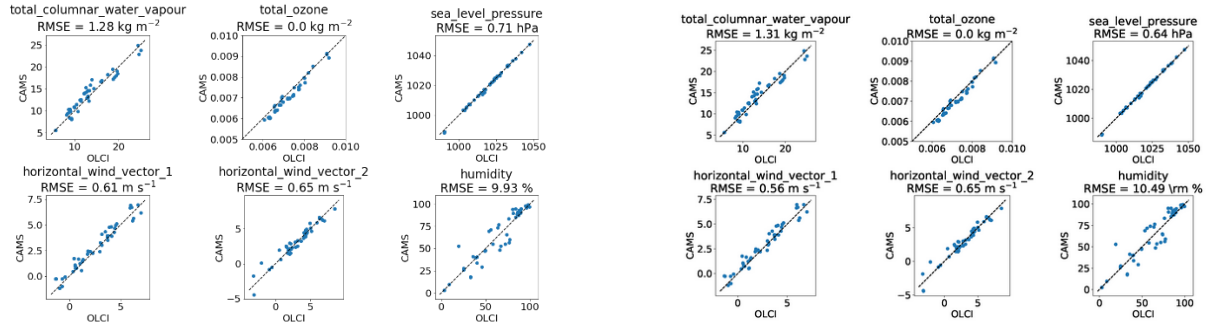


Figure 8: Figures seem to be equal.

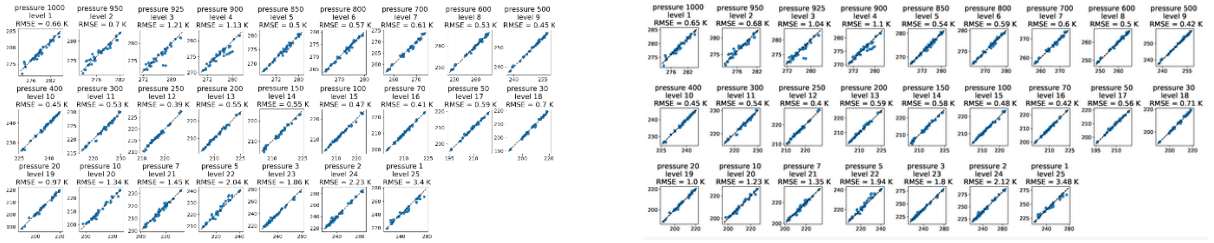


Figure 9: Figures seem to be equal

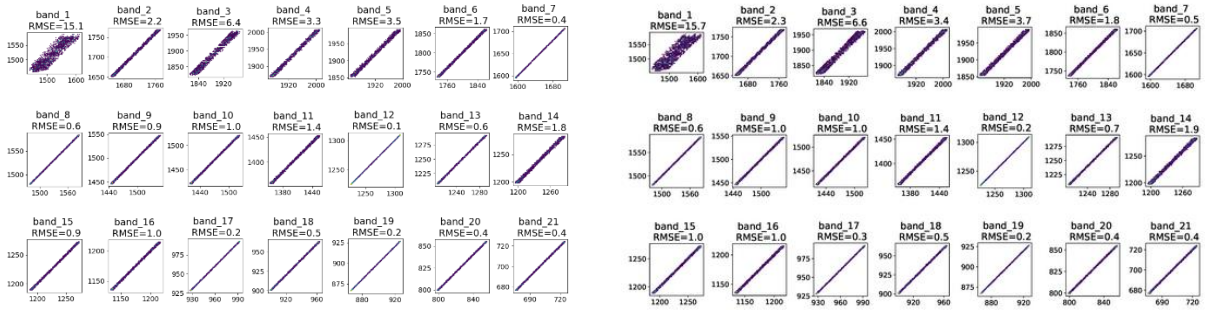


Figure A1: Figures seem to be equal. Deviations likely caused by differences in the scale.

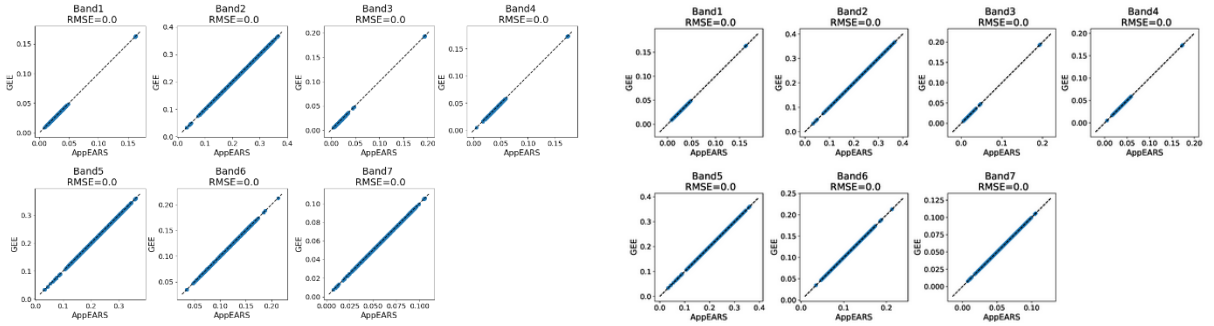
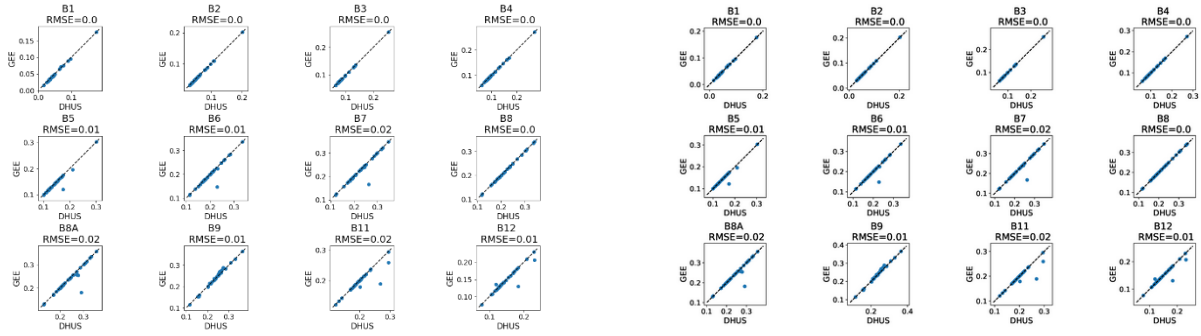


Figure A2: Figures seem to be equal. Deviations likely caused by differences in the scale.



Further comments

- At the time of publication, the CODECHECK was not yet available. For the next paper, I recommend doing a CODECHECK before submission.
- The GitHub repository does not contain the data, which is fine. However, the DANS repository could deposit the source code. This would make it possible to have one asset the includes all materials. This would also enable the use of tools such as Binder, which often can fetch materials from, for example, Dataverse (Note: DANS Easy will move their infrastructure to Dataverse soon).
- Mention in the README which python version was used.

Used computational environment

Windows 10 Enterprise, Intel Core i7-8665U CPU 1.90GHz 2.11GHz, 32GB RAM, 64-bit Operating System. Python 3.9.6, Jupyter 1.0.0, jupyterlab 3.0.16, netCDF 1.5.7, notebook 6.4.0, xarray 0.18.2